

BAYES' THEOREM

Q1. Bayes' Theorem Formula and Explanation

Definition

Bayes' Theorem is a statistical method used to find the probability of an event when we already have some related information or evidence.

It helps us **update our belief about something after receiving new information**.

Bayes' Theorem Formula

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

$$P(A|B) = \frac{0.85(0.20)}{0.25} = 0.68$$

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$$P(A)P(A)$$

$$P(A)P(A)$$

$$P(B|A)P(B|A)$$

$$P(B|A)P(B|A)$$

$$P(B|\neg A)P(B|\neg A)$$

$$P(B|\neg A)P(B|\neg A)$$

$$P(B)P(B) = 0.25$$

$$P(B \cap A)P(B \cap A) = 0.17$$

$$P(B \cap \neg A)P(B \cap \neg A) = 0.08$$

Give feedback

$$P(A|B) = \frac{P(B|A) \times P(A)}{P(B)}$$

Where:

Term	Meaning
$P(A)$	Prior Probability
$P(B A)$	Likelihood
$P(B)$	Evidence
$P(A B)$	Posterior Probability

1. Prior Probability — $P(A)$

The **prior probability** is what we believe about an event **before seeing the new evidence**.

Example:

Suppose from previous email records, 20% of emails are spam.

Therefore:

$$P(\text{Spam})=20\%$$

This is our **prior probability**.

2. Likelihood — $P(B|A)$

The **likelihood** tells us how likely the evidence is if the event is actually true.

Example:

If an email is spam, there may be a high chance that it contains words such as **"offer"**, **"free"**, **"win"**, etc.

So, the probability of seeing the word "offer" when an email is spam represents the **likelihood**.

3. Evidence — $P(B)$

The **evidence** is the overall probability of observing the evidence, regardless of whether the email is spam or not.

In this example, it means the probability that a randomly selected email contains the word **"offer"**.

4. Posterior Probability — $P(A|B)$

The **posterior probability** is the updated probability after considering the new evidence.

Example:

After seeing the word **"offer"**, we update our belief about whether the email is spam.

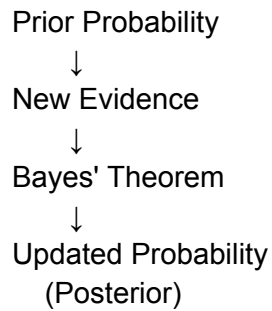
So:

$$P(\text{Spam} | \text{Offer})$$

means:

"What is the probability that the email is spam, given that it contains the word 'offer'?"

Simple Flow



Conclusion

Bayes' Theorem helps us make better probability-based decisions by combining our **previous knowledge (prior)** with **new evidence** to obtain an **updated probability (posterior)**.

Q2. Probability That an Email Containing "Offer" Is Spam

Given Data

Total emails = **1000**

Spam emails = **200**

Therefore, non-spam emails:

$$1000 - 200 = 800$$

The word **"offer"** appears in:

- 120 spam emails
- 80 non-spam emails

Step 1: Define the Events

Let:

S = Email is spam

O = Email contains the word "offer"

We need to find:

$$P(S | O)$$

That means:

Probability that an email is spam given that it contains "offer".

Step 2: Find the Prior Probability

There are 200 spam emails out of 1000 emails.

$$P(S) = \frac{200}{1000} \quad P(S) = 0.20 \quad \boxed{P(S) = 0.20}$$

or

$$P(S) = 20\% \quad \boxed{P(S) = 20\%}$$

Step 3: Find the Likelihood

The word "offer" appears in 120 spam emails out of 200 spam emails.

$$P(O|S) = \frac{120}{200} \quad P(O|S) = 0.60 \quad \boxed{P(O|S) = 0.60}$$

or

$$P(O|S) = 60\% \quad \boxed{P(O|S) = 60\%}$$

Step 4: Find the Evidence

The word "offer" appears in:

$$120 + 80 = 200$$

emails.

Therefore:

$$P(O) = \frac{200}{1000} \quad P(O) = 0.20 \quad \boxed{P(O) = 0.20}$$

or

$$P(O) = 20\% \quad \boxed{P(O) = 20\%}$$

Step 5: Apply Bayes' Theorem

Formula:

$$P(S|O) = \frac{P(O|S) \times P(S)}{P(O)}$$

Substitute the values:

$$P(S|O) = \frac{0.60 \times 0.20}{0.60 \times 0.20 + 0.12 \times 0.20} = \frac{0.12}{0.12 + 0.04} = \frac{0.12}{0.16} = 0.75$$

Therefore:

$$P(S|O) = 75\% \boxed{P(S|O) = 75\%}$$

Final Answer

The probability that an email containing the word "offer" is spam is 75%.

Verification Using Counts

There are **200 emails containing "offer"** in total:

$$120 \text{ spam} + 80 \text{ non-spam} = 200$$

Out of these 200 emails, 120 are spam:

$$\frac{120}{200} \times 100 = 60\%$$

So the answer is confirmed:

$$60\% \boxed{60\%}$$

Q3. COVID-19 Rapid Test Using Bayes' Theorem

Given

- Sensitivity = **98%**
- Specificity = **95%**
- Prevalence = **5%**

We need to find:

$$P(\text{COVID} | \text{Positive}) \boxed{P(\text{COVID} | \text{Positive})}$$

That means:

Probability that a person actually has COVID-19 given that their test result is positive.

Step 1: Define the Events

Let:

C = Person has COVID-19

P = Person tests positive

Step 2: Prior Probability

The percentage of people who actually have COVID-19 is 5%.

Therefore:

$$P(C)=0.05 \quad P(C)=0.05 \quad P(C)=5\% \quad \boxed{P(C)=5\%}$$

This is the **prior probability**.

Step 3: Likelihood

Sensitivity is 98%.

Sensitivity means:

Probability that the test is positive when the person actually has COVID-19.

Therefore:

$$P(P|C)=0.98 \quad P(P|C)=0.98 \quad P(P|C)=98\% \quad \boxed{P(P|C)=98\%}$$

Step 4: False Positive Probability

Specificity = 95%.

Specificity means:

$$P(\text{Negative} | \text{No COVID})=95\% \quad P(\text{Negative} | \text{No COVID})=95\%$$

Therefore, the false positive rate is:

$$100\% - 95\% = 5\% \quad 100\% - 95\% = 5\%$$

So:

$$P(P | \text{No COVID})=0.05 \quad P(P | \text{No COVID})=0.05$$

Step 5: Evidence — Probability of a Positive Test

We need:

$$P(P)P(P)$$

Using the total probability rule:

$$P(P) = P(P|C)P(C) + P(P|\text{No COVID})P(\text{No COVID}) \\ P(P) = P(P|C)P(C) + P(P|\text{No COVID})P(\text{No COVID})$$

Since:

$$P(\text{No COVID}) = 1 - 0.05 = 0.95 \quad P(\text{No COVID}) = 1 - 0.05 = 0.95$$

Therefore:

$$P(P) = (0.98)(0.05) + (0.05)(0.95) \\ P(P) = (0.98)(0.05) + (0.05)(0.95) \\ P(P) = 0.049 + 0.0475 \\ P(P) = 0.0965 \quad P(P) = 0.0965$$

Step 6: Apply Bayes' Theorem

$$P(C|P) = \frac{P(P|C) \times P(C)}{P(P)} \quad P(C|P) = \frac{P(P|C) \times P(C)}{P(P)}$$

Substitute the values:

$$P(C|P) = \frac{0.98 \times 0.05}{0.0965} \quad P(C|P) = \frac{0.98 \times 0.05}{0.0965} \\ P(C|P) = \frac{0.049}{0.0965} \quad P(C|P) = \frac{0.049}{0.0965} \quad P(C|P) = 0.5078 \quad P(C|P) = 0.5078$$

Converting into percentage:

$$0.5078 \times 100 = 50.78\% \quad 0.5078 \times 100 = 50.78\%$$

Final Answer

$$P(\text{COVID} | \text{Positive}) \approx 50.78\% \quad \boxed{P(\text{COVID} | \text{Positive}) \approx 50.78\%}$$

Therefore, **a person who tests positive has approximately a 50.78% probability of actually having COVID-19**, given the stated sensitivity, specificity, and 5% prevalence.

Important Observation

Even though the test has **98% sensitivity and 95% specificity**, the probability of actually having COVID-19 after a positive result is about **50.78%** because the disease prevalence is only **5%**.

This demonstrates why the **prior probability** is important in Bayes' Theorem.

Q4. Application of Bayes' Theorem in Gmail

Application: Spam Email Filtering

An application where Bayes' Theorem can improve user experience is **Gmail**.

Gmail can use probability-based methods to determine whether an incoming email is likely to be spam.

Scenario

Suppose a user receives an email containing words such as:

- "Congratulations"
- "You won"
- "Free"
- "Offer"
- "Click now"

These words can provide evidence that the email might be spam.

How Bayes' Theorem Can Help

The system can first consider the **prior probability**.

For example, based on previous email patterns, suppose the system estimates that a certain percentage of incoming emails are spam.

This becomes the **prior probability**.

When a new email arrives, the system examines evidence such as:

- Words used in the email
- Sender information
- Links
- Previous email patterns
- User's interaction history

The system then updates the probability using Bayes' Theorem.

Prior Probability

The probability that an incoming email is spam **before examining the new email's contents**.

$P(\text{Spam})$

Evidence

The presence of suspicious words or other spam-related characteristics.

For example:

Evidence="Free Offer" $Evidence = \text{"Free Offer"}$

Posterior Probability

After considering the evidence, the system calculates:

$P(\text{Spam} | \text{Evidence})$

This represents the updated probability that the email is spam.

Simple Process

Previous Email Information



Prior Probability



New Email Arrives



Examine Evidence



Bayes' Theorem



Posterior Probability



Spam / Not Spam Decision

Example

If the system initially believes that an email has a **20% chance of being spam**, this is the prior probability.

After detecting several suspicious characteristics, the probability may increase.

For example:

$P(\text{Spam})=20\%$

After considering the evidence:

$P(\text{Spam} | \text{Evidence})=85\%$

The system can then classify the email as **likely spam**.